FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Neonatal Tetanus

Tutorial | Week 6 · Tetanus, skin, poisons & hormones | Neonatology | Dr Nabbale

Neonatal tetanus is a toxin disease of babies born to unimmunised mothers whose umbilical cord was cut or dressed with contaminated material. A baby who sucked normally for the first 2 days stops sucking between days 3 and 28, develops trismus, rigidity and spasms while remaining conscious. Treat with human tetanus immunoglobulin, metronidazole, sedation, a quiet dark room and meticulous nursing; prevent it with maternal Td and clean cord care.

Must know before the ward round

1. WHO case definition - a baby who could suck and cry normally in the first 2 days of life, then loses the ability to suck between days 3 and 28 and becomes stiff (rigid) and/or has spasms.
2. Cause - spores of Clostridium tetani enter the umbilical stump (unsterile cutting instrument, cow dung, ash or herbs on the cord) in a baby whose mother has no tetanus antibodies (not immunised with Td/TT).
3. Pathophysiology - tetanospasmin travels up motor nerves to the spinal cord and brainstem and blocks release of the inhibitory transmitters GABA and glycine, causing rigidity, spasms and autonomic instability. The baby stays fully conscious.
4. Usually presents at 3-14 days (UCG) with inability to suck (trismus), irritability, excessive crying, then generalised rigidity, opisthotonus, clenched fists, risus sardonicus and spasms triggered by touch, noise or light.
5. Diagnosis is clinical - no laboratory test; CSF is normal. Short incubation (below 7 days) and short period of onset (below 48 h) mean worse prognosis.
6. Neutralise toxin - human tetanus immunoglobulin (TIG) 500 IU IM, split between 2 sites away from the vaccine site (UCG 2023); equine antitoxin if TIG is not available.
7. Kill the source - metronidazole 15 mg/kg IV loading over 60 min, then 7.5 mg/kg every 12 h (under 4 weeks old) (UCG); second line benzylpenicillin 100,000 IU/kg 12-hourly for 10-14 days; clean and debride the stump.
8. Control spasms - diazepam 0.2 mg/kg IV or 0.5 mg/kg rectally every 1-4 h as needed (UCG), plus chlorpromazine 1 mg/kg 8-hourly by NGT; phenobarbital is an alternative/adjunct; watch for apnoea.
9. Nursing - quiet, dark, cool room; minimal handling clustered after sedation; suction mouth; turn 30 min after sedative; IV fluids then NGT expressed breast milk from 24 h in small frequent feeds; monitor breathing and temperature.
10. Neonatal tetanus is a NOTIFIABLE disease in Uganda - report and investigate every case.
11. The disease does not give immunity - start the baby's routine DTP-containing vaccine (UCG - first dose of DPT at diagnosis) and complete the schedule; vaccinate the mother with Td now.
12. Prevention - Td for all women of childbearing age (Td2 at least 2 weeks before delivery; 5 doses for lifelong protection), clean delivery and cord care (chlorhexidine 7.1%, nothing else on the stump).

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology
5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup

9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Neonatal tetanus (tetanus neonatorum) is generalised tetanus in a baby in the first 28 days of life. It happens when **spores of *Clostridium tetani*** contaminate the **umbilical stump** (cut with an unsterile razor or reed, or dressed with cow dung, ash, soil or herbs) in a baby whose **mother has no protective antibodies** because she was not immunised with **tetanus toxoid (TT/Td)**.

The story is classic: a baby who **sucked and cried normally for the first 2 days**, then on day 3–14 **stops sucking** (jaw spasm, trismus), cries excessively, becomes **stiff**, and develops **painful spasms** triggered by touch, noise or light, while remaining **fully conscious**.

Term	Definition
Neonatal tetanus (WHO surveillance definition, confirmed case)	Illness in an infant with **normal ability to suck and cry in the first 2 days**, who **loses this ability between days 3 and 28** and **becomes rigid or has spasms**
Suspected case	Any neonatal death between 3 and 28 days of unknown cause, or any illness reported as tetanus in the first month, where the infant fed normally after birth
Maternal tetanus	Tetanus during pregnancy or within 6 weeks after the end of pregnancy (birth or abortion)
Incubation period	Time from spore entry to first symptom: usually **3–14 days** in neonates (often 5–7)
Period of onset	Time from first symptom to first spasm; **<48 h = severe**
MNT elimination	Fewer than **1 case per 1,000 live births in every district** per year (WHO target)

Other forms of tetanus (for comparison): generalised (commonest overall), localised (spasm near the wound), cephalic (cranial nerves, from head wounds or otitis).

> **CLINICAL PEARL**
> The one thing never to forget: **every case of neonatal tetanus is a failure of prevention**. When you admit one, also vaccinate the mother, report the case, and ask how the cord was cut and dressed.

2. Epidemiology

- **Global**: WHO reports that in **2021 just over 7,700 newborns died of neonatal tetanus**, an **84% reduction since 2000** (about 46,900 deaths), mainly due to immunisation of women with tetanus-toxoid-containing vaccine (TTCV) and clean birth practices. Disruption of immunisation services (for example during COVID-19) caused resurgence in some places.
- **Why it matters in Uganda**: neonatal tetanus is almost **100% preventable** with maternal Td and clean cord care, yet cases still occur after home births and harmful cord practices, especially in hard-to-reach districts. Without intensive care it is usually fatal; with good nursing, sedation and antitoxin, many babies survive.
- **Uganda status**: Td is given at ANC (first contact) and in the UNEPI schedule for girls and women of childbearing age. Uganda was validated by WHO as having achieved maternal and neonatal tetanus (MNT) elimination (2011), which must be sustained through Td coverage, facility births and clean cord care. Sporadic cases still present to regional and national referral hospitals.

- **Notifiable**: neonatal tetanus is a **notifiable disease** (UCG 2023). Notify the district surveillance focal person; every case triggers a case investigation (mother's Td history, place of birth, cord practices) and a community response.

Services by level of care Verify *compiled from the source notes, not an MoH equipment list*

Level	Usually available	Often missing
HC III / HC IV	Diazepam, benzylpenicillin, metronidazole (oral, sometimes IV), Td vaccine, pentavalent vaccine	TIG, oxygen and pulse oximetry at HC III, staff for continuous observation
General Hospital	TIG or equine antitoxin (variable), IV metronidazole, diazepam, chlorpromazine, NGT, IV fluids, oxygen concentrator	Neonatal ICU, ventilation
RRH	All the above, neonatal unit, blood culture at some, chest X-ray	Neonatal ventilation at most
NRH (Mulago) and a few private hospitals	Neonatal ventilation (limited beds), full laboratory	**ICU ventilation** for neonates is rare outside Kampala

3. Aetiology and Risk Factors

Organism: *Clostridium tetani*, an anaerobic, spore-forming Gram-positive bacillus found in **soil and animal faeces**. Spores survive for years and resist boiling and many antiseptics.

Portal of entry: the **necrotic umbilical stump** (an ideal low-oxygen environment); rarely circumcision wounds, ear piercing or scarification.

Risk factors

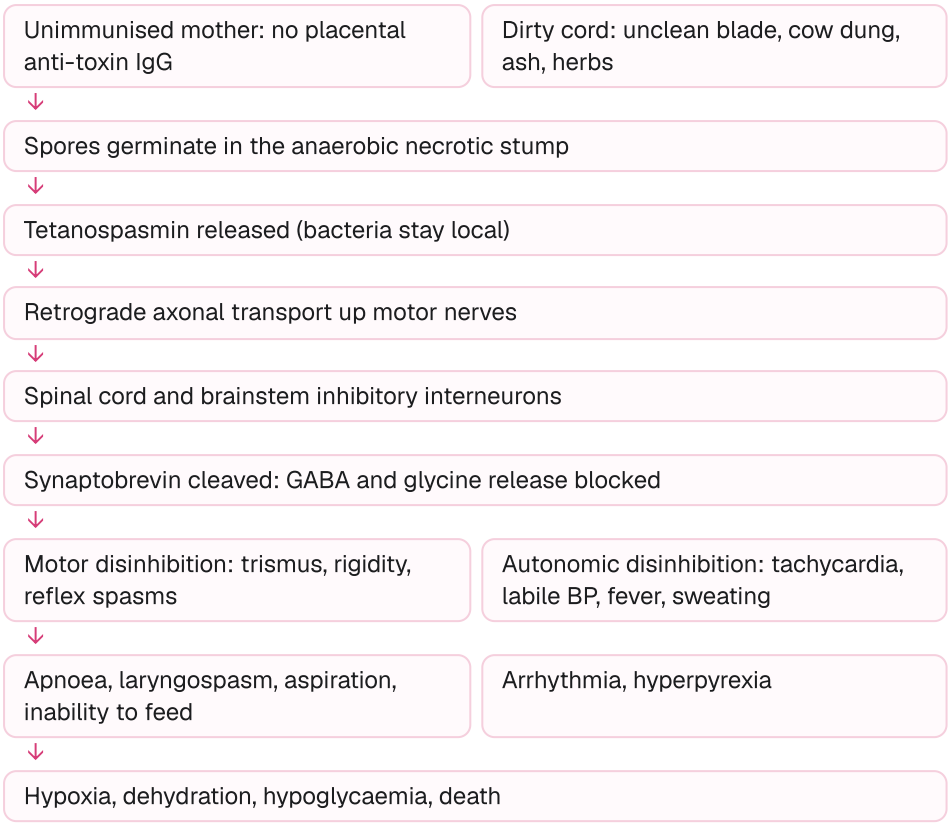
Mother (no antibodies)	Birth and cord (spore exposure)	Context
Unimmunised or incompletely immunised mother	**Home or roadside birth**, unskilled attendant	Rural residence, poverty
No ANC; first pregnancy without Td2 before delivery	**Unclean cutting instrument** (used razor blade, reed, knife)	Hard-to-reach districts
Teenage mother	Unclean cord tie	Cultural practices
Previous baby who died at 1–2 weeks (likely tetanus)	**Substances on the cord**: cow dung, ash, mud, herbs, ghee, powder	Disrupted immunisation services

4. Pathophysiology

Causal chain

1. **Spores** of *C. tetani* enter the **necrotic umbilical stump**, an ideal low-oxygen environment.
2. Spores germinate and the bacteria produce **tetanospasmin** (and tetanolysin). The bacteria stay local; the **toxin** causes the disease.
3. Tetanospasmin binds to **motor nerve terminals** and travels by **retrograde axonal transport** to the **spinal cord and brainstem**.
4. In the central nervous system it is taken up by **inhibitory interneurons** and cleaves **synaptobrevin**, blocking the release of the inhibitory neurotransmitters **GABA and glycine**.
5. Without inhibition, motor neurons fire excessively → **rigidity** and **reflex spasms**. The shortest nerves (cranial: jaw, face) are affected first → **trismus**, **risus sardonicus**, then neck, back (**opisthotonus**), abdomen and limbs.

6. **Autonomic** disinhibition → tachycardia, labile blood pressure, sweating, hyperpyrexia, arrhythmias.
7. **Toxin binding is irreversible**: recovery needs **new nerve terminal sprouting**, which takes **3–6 weeks**. Antitoxin only neutralises **unbound** toxin, so give it **early**.
8. **Why the mother matters**: maternal IgG antibodies from TT/Td cross the placenta and neutralise toxin in the baby. That is why maternal immunisation prevents neonatal tetanus.
9. **Why the disease does not protect**: the amount of toxin that causes disease is too small to induce immunity, so survivors must be vaccinated.



Feature links

- Shortest motor pathways (cranial nerves) disinhibited first → masseter spasm → **trismus** → baby cannot suck (the first symptom the mother notices).
- Facial muscle spasm → **risus sardonicus** (fixed grimace), wrinkled forehead, pursed lips ("carp mouth").
- Paraspinal muscle spasm → **opisthotonus**.
- Abdominal wall rigidity → **board-like abdomen**.
- Limb flexor and extensor co-contraction → **clenched fists**, flexed arms, extended legs, dorsiflexed toes.
- Loss of reflex inhibition → **spasms provoked by touch, noise, light, injection**.
- Toxin spares the cortex → baby is **conscious and alert between spasms** (and feels pain).
- Spasm of laryngeal and respiratory muscles → **apnoea, laryngospasm, cyanosis**.
- Trismus and pharyngeal spasm → secretions pool → **aspiration pneumonia**.
- Cannot suck → **dehydration, hypoglycaemia, weight loss**.
- Autonomic storm → **tachycardia >180/min, labile BP, sweating, fever**.
- Muscle activity of repeated spasms → **fever** (even without infection), exhaustion, rarely **vertebral fractures**.
- Hypoxia during spasms → later **developmental delay or cerebral palsy** in some survivors.
- Irreversible toxin binding → **slow recovery over 3–6 weeks**, regardless of antitoxin.

5. Classification / Staging

There is no single bedside staging score in UCG 2023. Classify by **case definition** (section 1) and by **severity markers**, which also predict prognosis.

Marker	Milder	Severe
Incubation period	≥7 days	**<7 days**
Period of onset (first symptom to first spasm)	≥48 h	**<48 h**
Spasms	Infrequent, short, only on handling	**Frequent, prolonged, spontaneous**, with cyanosis or apnoea
Autonomic features	Absent	**Fever, tachycardia >180, labile BP**
Baby	Term, normal weight	**Preterm or LBW**

For the ward, record a **spasm chart** (spasms per hour, duration, cyanosis or apnoea) every shift: it is your working severity score and guides sedation.

6. Clinical Features

History

Ask the mother gently: she may feel blamed. Each question has a reason.

Question	Why it matters
Age now and day symptoms began	Onset day 3–28 fits; shorter incubation (<7 days) = worse prognosis
Did the baby suck and cry normally in the first 2 days?	Key part of the WHO case definition (separates from HIE and congenital problems)
When did the baby stop sucking? First spasm when?	Period of onset <48 h = severe
Place of birth, who delivered, what was used to cut and tie the cord	Unclean birth is the source
What was applied to the cord? (cow dung, ash, herbs, ghee, powder)	Source of spores
Mother's Td/TT doses and timing (Td card), ANC attendance	Unprotected mother; plan her vaccination
Previous babies who died at 1–2 weeks	Suggests previous tetanus in an unimmunised mother
Feeding since onset: any fluids, how given? Urine output	Dehydration, hypoglycaemia, aspiration risk
Breathing pauses, colour change during spasms	Apnoea, laryngospasm: life-threatening
Fever, convulsions with loss of consciousness, jaundice	Points to sepsis, meningitis or other diagnoses
Maternal HIV status, syphilis	Other risks; prophylaxis
Baby's immunisations (BCG, OPV0)	Usually missed after home birth; give before discharge

Examination

Follow Hutchison's Clinical Methods (24th edition) order adapted for the neonate, but **observe first, touch later**: handling triggers spasms. If spasms are ongoing, **sedate first** (section 9) and cluster the examination about 30 minutes after the sedative dose.

Introduction, consent and preparation

- Introduce yourself to the mother; confirm the baby's name, age in days and sex; explain that you will look first and touch as little as possible; gain consent.
- **Hand hygiene** before touching the baby.
- **Position and exposure**: examine in the cot or on a warm surface in a **quiet, dim** area; expose only the part you are examining and re-cover quickly (avoid hypothermia and stimulation).

General inspection (end of the cot, before touching)

Sign	Mechanism	What it means
Posture: clenched fists, flexed arms, extended legs, dorsiflexed toes	Generalised rigidity from loss of inhibition	Generalised tetanus
Spasms (count frequency and duration; note cyanosis or apnoea)	Reflex motor discharge	Severity; titrate sedation
Alert, anxious face between spasms	Cortex spared	**Key difference from seizures and meningitis**
Opisthotonus during spasms	Paraspinal spasm	Generalised tetanus (also kernicterus: check jaundice)
Colour: cyanosis, jaundice, pallor	Apnoea or laryngospasm; kernicterus; sepsis	Danger signs
Hydration and nutrition: sunken fontanelle and eyes, dry mouth, wasting	Unable to suck	Dehydration, weight loss

Vital signs and growth: temperature (fever from spasms or infection; or hypothermia), HR (tachycardia, autonomic instability), RR (count a full minute), SpO_2, BP if possible, capillary refill; **weight** daily (dehydration, fluid calculation), compared with birth weight if known.

Hands, face and neck

- **Hands**: clenched fists; peripheral cyanosis; capillary refill.
- **Face**: **risus sardonicus** (fixed grimace), wrinkled forehead, pursed lips ("carp mouth").
- **Mouth: trismus**: the mouth cannot open; baby cannot suck. Try to insert a finger or spatula gently (**spatula test**: the baby bites down reflexively in tetanus instead of gagging). Do this last and only once: it provokes spasms.
- **Neck**: neck stiffness from generalised rigidity (does not by itself mean meningitis).

Nervous and musculoskeletal system

- **Inspection**: posture, spontaneous spasms, opisthotonus.
- **Palpation (tone)**: **generalised rigidity**, **board-like abdomen**, stiff limbs; spasms provoked by touch, noise, light or injection.
- **Level of consciousness**: **alert between spasms**.
- **Fontanelle**: normal (bulging suggests meningitis; sunken suggests dehydration).

Completing the examination

- **Umbilicus**: dirty, infected or septic stump (it may look normal by the time of presentation); flare of redness around the stump (omphalitis).

- **Chest**: crackles of **aspiration pneumonia**; signs of sepsis.
- **Skin**: jaundice, pressure areas, injection sites.
- **Thank** the mother, re-cover the baby, **hand hygiene**, and **summarise** with a severity assessment and plan (model presentation in section 8).
- Document the **spasm frequency** each shift to guide sedation.

Red flags

RED FLAG

- **Apnoea, cyanosis or laryngospasm** during spasms
- **Very frequent or prolonged spasms** despite sedation
- **Short incubation (<7 days)** or **period of onset <48 h**
- **Fever, tachycardia >180, labile BP** (autonomic instability)
- **Aspiration pneumonia**, sepsis
- **Over-sedation**: no spontaneous breathing, floppy, SpO_2 falling after diazepam
- **Hypoglycaemia, hypothermia, dehydration** from not feeding

Thinking like a clinician

- **Conscious baby + spasms + trismus + onset day 3–14 + dirty cord = tetanus.** A baby with seizures from meningitis or hypoglycaemia has altered consciousness between episodes; the tetanus baby looks anxious and alert.
- **"Could suck at birth, cannot suck now"** separates tetanus from HIE (abnormal from birth) and from congenital problems.
- **Do not do things that provoke spasms** without sedation (LP, NGT insertion, IV line, examination): sedate first and cluster procedures.
- **The biggest killers are apnoea, aspiration and exhaustion**, not the toxin itself. Good nursing saves lives.
- **Sedation is a balance**: enough to stop spasms, not so much that the baby stops breathing. Know where the bag and mask are.
- **Always ask about the mother's Td** and vaccinate her before she leaves.
- **If X then think Y**: opisthotonus with deep jaundice → kernicterus; jitteriness with carpopedal spasm and no trismus → hypocalcaemia; bulging fontanelle with drowsiness → meningitis; abnormal from birth after a difficult labour → HIE.
- **Common student mistakes**: treating the "convulsions" with phenobarbital alone; bottle-feeding a baby with trismus; giving TIG and vaccine into the same limb; forgetting to notify and to vaccinate the mother.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Neonatal sepsis / meningitis	Fever or hypothermia, lethargy, **reduced consciousness**, bulging fontanelle, seizures not triggered by touch	Blood culture, LP (abnormal CSF)
Hypocalcaemia (neonatal tetany)	Jitteriness, carpopedal spasm, seizures; often infant of a diabetic mother, preterm or asphyxiated; **no trismus**	Serum calcium
Hypoglycaemia	Jittery, lethargic, seizures; **responds to glucose**	Glucose
HIE seizures	**Abnormal from birth** after a difficult delivery; altered consciousness	History, Sarnat staging

Differential	Distinguishing clinical feature	Distinguishing investigation
Kernicterus	**Deep jaundice**, opisthotonus, high-pitched cry, lethargy	Bilirubin
Neonatal abstinence / drug withdrawal	Maternal drug use; jittery, hypertonia, poor feeding, sneezing	History
Congenital neuromuscular or structural problems	Present from birth; no spasms	Clinical, imaging

Neonatal tetanus itself: normal suck days 1–2, then trismus, rigidity, spasms provoked by stimuli, **conscious** between spasms, dirty cord; the diagnosis is clinical.

8. Clinical Workup

No test is diagnostic: tetanus is a clinical diagnosis. Treat on clinical grounds. Tests look for complications and mimics. "Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes*.

Bedside

Test	Why	Expected finding	Available at
Glucose	Hypoglycaemia from not feeding (mimic and complication)	Low if not fed; normal in tetanus itself	HC III and above (glucometer)
Temperature	Fever (spasms, infection) or hypothermia	Often mildly raised from muscle activity	All levels
SpO_2	Hypoxaemia during spasms or apnoea; over-sedation	Falls during spasms	HC IV and above; often missing at HC III
Weight	Dehydration; fluid and drug calculation	Weight loss	All levels
Spasm chart	Number and duration of spasms per hour to titrate sedation	Falling count with effective sedation	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	Baseline, suspected sepsis	Often normal; leucocytosis if secondary infection	Antibiotic choice	HC IV and above

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Chest X-ray	Suspected aspiration pneumonia	Consolidation	Antibiotics, oxygen	Most hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Blood culture	Fever, umbilical sepsis	Secondary bacterial sepsis (*S. aureus*, Gram-negatives)	Add or adjust antibiotics	Referral hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
Electrolytes, calcium	Doubt about the diagnosis	Hypocalcaemia (tetany mimic)	Calcium gluconate	Referral hospitals
Lumbar puncture (after sedation)	Only if meningitis cannot be excluded clinically	**Normal CSF** in tetanus	Meningitis needs 21 days of antibiotics	Hospital
Wound swab culture	Rarely useful	*C. tetani* is hard to grow and may be present without disease	Not used to diagnose	Limited

Putting it together

Model one-line summary

> "A **9-day-old** male born at **home in Kayunga** by a traditional birth attendant, cord cut with a **used razor blade** and dressed with **cow dung**, who **sucked well until day 6**, then **stopped sucking**, and since day 7 has **generalised spasms** triggered by touch with **trismus, risus sardonicus and opisthotonus**, alert between spasms. Mother is a **19-year-old primigravida** who attended ANC once and received **one dose of Td**. Weight 2.8 kg, T 38.0 °C, HR 172, SpO_2 92% in air."

Assessment: **neonatal tetanus**, incubation 6 days and period of onset about 24 h (**severe**), with possible omphalitis; risks of apnoea, aspiration and dehydration.

Problem list

1. Neonatal tetanus with frequent spasms.
2. Source: infected umbilical stump.
3. Feeding and hydration.
4. Risk of apnoea and aspiration.
5. Unprotected mother.
6. Notification and community follow-up.

Model case presentation (ward round)

> "**Baby Kato**, a **10-day-old** male, weight **2.9 kg** (birth weight not known; home birth), **admitted yesterday with neonatal tetanus**. He breastfed normally until **day 6**, then stopped sucking; spasms began on **day 7**. The cord was cut with a **razor blade** and **cow dung** was applied. Mother is 20, had **one dose of Td** in this pregnancy, **HIV-negative** at ANC, RPR negative. He has had no BCG or OPV0.
>
> On examination he has **trismus, risus sardonicus, clenched fists and opisthotonic spasms** provoked by touch, **alert between spasms**; T 37.8 °C, HR 168, RR 52, SpO_2 95% in 1 L/min oxygen; umbilicus moist with surrounding redness 0.5 cm.
>
> He has received **TIG 500 IU IM in two sites**, the **first pentavalent dose**, **metronidazole 43.5 mg loading then 22 mg 12-hourly**, and is on **diazepam 0.6 mg IV 2-hourly** alternating with **chlorpromazine 3 mg 8-hourly by NGT**. Spasms have reduced from **12 to 5 per hour**. He is on IV fluids at 120 mL/kg/day and started **EBM 5 mL 2-hourly** by NGT today.
>
> **Problems:** severe neonatal tetanus (incubation 6 days, onset to spasm 1 day); umbilical infection; feeding; risk of apnoea and aspiration; unimmunised mother. **Plan:** continue sedation titrated to spasms, cluster care, advance NGT feeds, monitor SpO_2 and apnoea, **Td for the mother today**, **notify** the district surveillance officer."

Worked example: doses for a 2.8 kg, 9-day-old baby (UCG doses)

DRUG DOSING

TIG **500 IU IM** (250 IU into each thigh, separate from the vaccine site); first pentavalent (DTP-containing) dose in another site; **metronidazole 42 mg IV** over 60 min, then **21 mg IV 12-hourly**; **diazepam 0.56 mg IV** (or **1.4 mg rectally**) every 1–4 h as needed; **chlorpromazine 2.8 mg via NGT 8-hourly**.

9. Management

The five pillars

1. **Airway, breathing and spasm control** (sedation, suction, oxygen, ventilation if available).
2. **Neutralise unbound toxin** (TIG or antitoxin, as early as possible).
3. **Remove the source** (metronidazole; clean and debride the stump).
4. **Supportive and nursing care** (quiet dark room, fluids, NGT feeding, temperature, skin, eyes, monitoring).
5. **Prevention**: vaccinate the baby (and complete the schedule) and the **mother**, **notify**, and educate.

Immediate / emergency

First 5 minutes (ABCDE)

- **A**: suction the mouth gently; position on the side.
- **B**: oxygen if cyanosed or SpO_2 <90%; **bag-mask ventilation** if apnoeic.
- **C**: HR, capillary refill; **IV access after sedation**; start IV fluids.
- **D**: **stop the spasm**: **diazepam 0.5 mg/kg rectally** or **0.2 mg/kg IV** (UCG) if spasms are ongoing; watch breathing. **Glucose** (when sedated).
- **E**: temperature; **quiet, dark, cool corner or room**; minimal handling.
- **Call for help**; plan TIG, metronidazole and nursing.
- **At HC III/IV**: sedate, give the first dose of antibiotic (metronidazole or benzylpenicillin) and TIG if available, and **refer immediately** to hospital (UCG: "Refer to hospital immediately") with the mother.

Definitive

- **Neutralise toxin**: TIG 500 IU IM in two sites, as early as possible.
- **Remove the source**: metronidazole (first line) for the UCG course; clean and **debride** the stump; chlorhexidine; nothing else on the cord.
- **Control spasms**: diazepam titrated to the spasm chart, with chlorpromazine via NGT to reduce the total benzodiazepine dose; phenobarbital as an alternative or adjunct.
- **Vaccinate**: first DTP-containing dose at diagnosis in a different limb; mother's Td before she leaves.
- **Supportive care**: the nursing bundle below is the most important part.

Drug table

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
Human tetanus immunoglobulin (TIG)	**500 IU** (fixed dose)	**IM**, divided between **at least 2 sites**, away from the vaccine site	**Once**	Single dose	Neutralises unbound toxin only: give as early as possible. Intrathecal TIG has been studied in neonatal tetanus but is **not routine**

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
Equine tetanus antitoxin (ATS) (if TIG unavailable)	**5,000 IU** (WHO and older protocols; check local product) Verify *dose against product insert and current UCG*	IM (or IV per product)	Once	Single dose	Test dose and **adrenaline ready** (anaphylaxis, serum sickness)
DTP-containing vaccine (UCG: "1st dose of DPT"; in Uganda the pentavalent DTP-HepB-Hib)	Standard dose	IM, **different limb** from TIG	At diagnosis	Continue the routine schedule	Disease does **not** confer immunity
Metronidazole (first line)	**Loading 15 mg/kg** over 60 min, then **7.5 mg/kg**	IV (oral or NGT when possible)	**Every 12 h if <4 weeks old**; **every 8 h if >4 weeks**	UCG 2023 lists a **14-day** course (many references use 7–10 days)	Preferred over penicillin
Benzylpenicillin (second line)	**100,000 IU/kg**	IV/IM	**Every 12 h**	**10–14 days** (UCG)	Metronidazole is preferred (penicillin is a GABA antagonist)
Diazepam (spasm control)	**0.2 mg/kg IV** or **0.5 mg/kg rectal**	IV / rectal	**Every 1–4 h** as needed (UCG)	Taper slowly over weeks as spasms settle	Titrate to spasms; **apnoea**, hypotension; some units use a continuous infusion with monitoring
Chlorpromazine (adjunct)	**1 mg/kg**	Oral via **NGT**	**Every 8 h** (UCG)	Until spasms settle	Often alternated with diazepam to reduce total benzodiazepine dose; hypotension
Phenobarbital (alternative or adjunct sedative)	Loading **20 mg/kg**, then **5 mg/kg/day**	IV / IM / NGT	Daily	Per local protocol	Additive respiratory depression
Magnesium sulfate	Specialist or ICU use only	IV	Infusion	n/a	Reduces spasms and autonomic instability in adults; limited neonatal evidence
Cloxacillin + gentamicin	Cloxacillin **50 mg/kg** + gentamicin **5 mg/kg** (7.5 mg/kg after week 1)	IV	Cloxacillin 6-hourly; gentamicin once daily	Per response	Only if **omphalitis with flare** or secondary sepsis
Oxygen	To SpO_2 >90%	Nasal prongs	As needed	During spasms and apnoea	Bag and mask at the bedside
Td (mother)	0.5 mL	IM	Today, then complete the 5-dose schedule	5 doses	Give before she leaves the ward

UGANDA CONTEXT

UCG 2023 places neonatal tetanus management at **hospital level (H)** and ventilation at **regional/national referral level (RR)**. If TIG is out of stock, use equine antitoxin with anaphylaxis precautions, or IV immunoglobulin (which contains tetanus antitoxin) where available. Do not delay sedation, metronidazole and nursing while you look for TIG. Metronidazole course: UCG 14 days versus 7–10 days in many international references; follow UCG in Ugandan practice and exams.

Referral criteria

Level	What to do	Refer up when
Community / VHT	Recognise a baby who stops sucking with stiffness after day 2	Immediately, with the mother
HC III / HC IV	Sedate (diazepam rectal or IV), first dose of metronidazole or benzylpenicillin, TIG if available, keep warm, suction	**Every case**: "Refer to hospital immediately" (UCG) with the mother
General Hospital (UCG level H)	Full five-pillar management, nursing bundle, NGT feeds	Recurrent apnoea, uncontrollable spasms despite maximal sedation, need for ventilation → RRH
RRH	All the above; neonatal unit; ventilation in some	Ventilation unavailable → NRH
NRH (Mulago) and a few private hospitals	Neonatal ventilation, ICU	n/a

Referral checklist: sedated before transfer; airway suctioned; a health worker who can bag-ventilate travels with the baby; mother accompanies (breast milk, history, Td); referral note with drug doses and times.

Monitoring and follow-up

- **Monitoring chart**: spasms per hour, sedation doses, RR, HR, SpO_2, temperature, intake and output, weight daily.
- **Observe breathing continuously after diazepam**; bag and mask at the bedside.
- **Wean sedation slowly** over 2–4 weeks as spasms reduce; the baby may stay rigid for weeks; full recovery of sucking may take 2–4 weeks.
- **Discharge** when sucking well, no spasms and gaining weight; give BCG and OPV0 if not given and continue the UNEPI schedule.
- **Follow-up**: immunisation clinic for the remaining DTP-containing doses; developmental follow-up for survivors with hypoxic episodes (risk of developmental delay or cerebral palsy); mother's remaining Td doses at ANC or the immunisation clinic.

Nursing care (the most important part)

1. **Environment**: a **quiet, dark, cool** room or corner (UCG); cover the cot; no radios; restrict visitors (but keep the mother close).
2. **Minimal handling**: **cluster care** (feeds, observations, injections, turning) **about 30 minutes after a sedative dose** (UCG: suction and turn 30 min after sedative).
3. **Airway**: keep a **mucus extractor or suction** at the bedside; suction the mouth as needed; position on the side.
4. **Breathing**: oxygen and pulse oximetry; **bag and mask at the bedside** for apnoea; observe breathing continuously after diazepam.

5. **Fluids** (UCG): **start with IV fluids** (UCG: half-strength saline with 5% dextrose; follow neonatal fluid volumes by weight and day of life, for example 120–150 mL/kg/day after the first days), then **NGT feeds of expressed breast milk from 24 h after admission, in small frequent feeds**; reduce IV as feeds increase.
6. **Mother expresses breast milk** every 3 h to keep lactation going; she can do gentle care when sedation allows.
7. **Temperature**: monitor 4-hourly; treat fever by environmental measures; avoid hypothermia.
8. **Cord care**: clean and **debride** the stump; chlorhexidine; nothing else.
9. **Skin, eyes, mouth**: turn 2–3-hourly (pressure areas), eye lubrication if lids do not close, mouth care.
10. **Monitoring chart** as above.

Public health actions

- **Notify** every case (and any neonatal death on day 3–28 of unknown cause) to the district surveillance focal person.
- **Case investigation**: mother's Td history, place of birth, birth attendant, cord practices.
- **Community response**: Td catch-up for women of childbearing age in the area; education on clean delivery and cord care.

10. Complications

Disease-related

Acute	Later
Apnoea, laryngospasm, respiratory failure (main cause of death)	Most survivors recover fully
Aspiration pneumonia	Some have **neurological sequelae** from hypoxia during spasms (developmental delay, cerebral palsy)
Hypoglycaemia, dehydration, malnutrition	Recurrence possible without immunisation
Secondary sepsis (umbilical, line, hospital-acquired)	Family grief; future pregnancies at risk if the mother is not vaccinated
Autonomic instability: tachycardia, hyperthermia, arrhythmia	
Fractures (vertebral) from violent spasms (rare)	
Death: high case fatality, especially with short incubation and no ventilation	

Treatment-related

Complication	Cause	Recognition and action
Over-sedation, apnoea	Diazepam, phenobarbital, chlorpromazine (additive)	No spontaneous breathing, floppy, SpO_2 falling: bag-mask ventilate; reduce or space doses
Hypotension	Diazepam, chlorpromazine	BP monitoring; reduce dose
Anaphylaxis, serum sickness	Equine antitoxin	Test dose; adrenaline ready
Aspiration	Feeding by bottle or cup with trismus	IV fluids first, then NGT EBM
Line infection, hospital-acquired sepsis	Prolonged IV access	Asepsis; remove lines when feeds established

11. Prognosis

- **Poor prognostic features**: incubation **<7 days**, period of onset **<48 h**, frequent severe spasms, fever, tachycardia, LBW or prematurity, apnoea, no access to ventilation.
- Without intensive care the disease is usually fatal; with good nursing, sedation and antitoxin, many babies survive.
- Recovery is slow (**3–6 weeks**, the time to sprout new nerve terminals); sucking returns over 2–4 weeks.
- Most survivors recover fully; some have neurological sequelae from hypoxia during spasms.
- Survival does **not** give immunity.

12. Prevention

Level	Action
Mother (now)	Give **Td** before discharge (start or continue the 5-dose schedule)
Women of childbearing age	**Td1** at first contact; **Td2** at least 4 weeks later; **Td3** 6 months later; **Td4** 1 year later; **Td5** 1 year later (lifelong protection). Give **Td2 at least 2 weeks before delivery**
Childhood	DTP-containing vaccine (pentavalent) at **6, 10 and 14 weeks** (UNEPI), plus boosters per the current national schedule so that girls arrive at childbearing age protected
Birth	Facility delivery or a skilled attendant; the "**cleans**": clean hands, clean perineum, clean delivery surface, **clean blade**, **clean cord tie**, **clean cord care**
Cord care	**Chlorhexidine 7.1% digluconate gel daily for 7 days** (UCG), keep dry and clean, **nothing else on the stump**
Surveillance	**Notify** every case (and any neonatal death on day 3–28 of unknown cause), case investigation, community Td response

> **UGANDA CONTEXT**
> Uganda's ANC Td schedule is the same one in your Obs & Gyn quick reference (Td1 at first contact, Td2 4 weeks later, Td3 6 months later, Td4 and Td5 a year apart). When you see a mother with a tetanus baby, give her **Td today** and explain that she needs the remaining doses to protect future babies.

13. Special Populations

- **Preterm and LBW babies**: worse prognosis; more prone to apnoea with sedatives, hypoglycaemia and hypothermia; use preterm fluid volumes and monitor SpO_2 continuously.
- **Infants older than 4 weeks**: metronidazole interval changes to **every 8 h** (UCG).
- **HIV-exposed infants**: continue PMTCT prophylaxis and early infant diagnosis as per the national schedule; no change to tetanus treatment.
- **The mother**: she may herself be at risk (maternal tetanus up to 6 weeks after birth or abortion); examine her if she has jaw stiffness or a septic perineal wound, and vaccinate her.

14. Key Points

High-yield points

- WHO case definition: normal suck and cry for 2 days, then loss of suck between days 3 and 28, with rigidity or spasms.

- Conscious baby + trismus + spasms provoked by touch + dirty cord = tetanus.
- Diagnosis is clinical; CSF is normal.
- TIG 500 IU IM in two sites, away from the vaccine.
- Metronidazole first line; benzylpenicillin second line.
- Diazepam plus chlorpromazine, titrated to a spasm chart; bag and mask at the bedside.
- Quiet, dark, cool room; cluster care 30 minutes after sedation.
- IV fluids first, then NGT expressed breast milk from 24 h.
- Notify; vaccinate the baby and the mother.
- Prevention: maternal Td and the six cleans, chlorhexidine on the cord.

Numbers & doses

Item	Value
Case definition	Normal suck days **1–2**; lost between days **3–28**; rigidity or spasms
Usual onset (UCG)	**3–14 days**
Poor prognosis	Incubation **<7 days**; onset-to-spasm **<48 h**
TIG	**500 IU IM** in **2 sites** (UCG)
Equine antitoxin	**5,000 IU** (if TIG unavailable; anaphylaxis precautions)
Metronidazole	**15 mg/kg** load over 60 min, then **7.5 mg/kg 12-hourly** (<4 weeks)
Benzylpenicillin (second line)	**100,000 IU/kg 12-hourly** for **10–14 days**
Diazepam	**0.2 mg/kg IV** or **0.5 mg/kg rectal** every **1–4 h**
Chlorpromazine	**1 mg/kg 8-hourly** via NGT
NGT feeds	EBM from **24 h** after admission, small and frequent
Recovery time	**3–6 weeks** (new nerve terminals)
Td schedule	Td1 first contact; Td2 **+4 weeks**; Td3 **+6 months**; Td4 **+1 year**; Td5 **+1 year**
Protection at birth	**Td2 at least 2 weeks before delivery**
Global deaths	About **7,700 in 2021**, down **84%** since 2000 (WHO)
Notifiable?	**Yes** (UCG 2023)

Exam traps

- **"Baby with spasms on day 7: treat as neonatal seizures with phenobarbital only."** → Think **tetanus** if the baby is conscious, has trismus and spasms provoked by touch; give TIG, metronidazole, sedation.
- **"Do an LP and insert an NGT straight away."** → Sedate first and cluster procedures; stimulation triggers spasms and apnoea.
- **"Penicillin is the antibiotic of choice."** → **Metronidazole** is first line (UCG); penicillin is second line.
- **"The baby survived, so he is now immune."** → No. Give DTP-containing vaccine and complete the schedule.
- **"Treat the baby and discharge the mother."** → Give the **mother Td**, notify, and educate about cord care.
- **"Inject TIG into the same limb as the vaccine."** → Use **different sites**.
- **"Feed by bottle as soon as he is admitted."** → He cannot suck and may aspirate: IV fluids first, then **NGT EBM** from 24 h.

Questions seniors ask

Viva questions

1. *What is the WHO case definition?* Normal suck and cry in the first 2 days, then loss of ability to suck between days 3 and 28, with rigidity or spasms.
2. *How does tetanus toxin cause spasms?* It travels retrogradely to the CNS and blocks release of GABA and glycine from inhibitory interneurons.
3. *Why is the baby conscious?* The toxin affects inhibitory motor pathways, not the cortex.
4. *What is the dose of TIG?* 500 IU IM, divided into two sites, away from the vaccine (UCG).
5. *Why metronidazole rather than penicillin?* Equal or better activity against *C. tetani*, and penicillin is a GABA antagonist; metronidazole is associated with lower mortality.
6. *How would you prevent neonatal tetanus?* Maternal Td (5 doses; Td2 at least 2 weeks before delivery), clean delivery, clean cord care with chlorhexidine.
7. *Does surviving tetanus give immunity?* No. Vaccinate the baby and complete the schedule.

On the ward round

1. **"How do you know this is tetanus and not meningitis?"** Normal suck days 1–2, trismus, spasms triggered by stimuli, alert between spasms, normal fontanelle; meningitis has reduced consciousness, fever, bulging fontanelle.
2. **"What were the incubation period and period of onset?"** Give the days and hours; short = severe.
3. **"What has he received, and in what dose?"** TIG 500 IU, metronidazole mg/kg, diazepam mg/kg and frequency.
4. **"How are you titrating sedation?"** By spasm frequency and breathing; bag and mask at the bedside.
5. **"How is he being fed?"** IV fluids then NGT EBM; mother expressing.
6. **"What about the mother?"** Td today and the full schedule; ANC for future pregnancies.
7. **"Who have you told?"** Notified district surveillance; case investigation.

Mnemonics

Clinical features: "STIFF"

- **S**pasms provoked by touch, light, noise
- **T**rismus (cannot suck)
- **I**ntact consciousness
- **F**ists clenched, face grimacing (risus sardonicus)
- **F**ourth day onwards (days 3–28) after a normal start

Treatment: "TIG-MET-DIAZ-DARK-Td": TIG, metronidazole, diazepam, dark quiet room, Td for mother (and DTP for baby).

Six cleans of delivery: hands, perineum, surface, blade, tie, cord care.

OSCE tips

- Expect a **counselling station** on cord care or Td for a pregnant woman, or a **spot diagnosis** from a picture of opisthotonus or risus sardonicus.
- In a history station, always ask: "**Could the baby suck after birth?**", "**What was used to cut the cord, and what was put on it?**", "**How many tetanus injections did you get?**"
- In an examination station, say that you would **observe before touching**, sedate before procedures, and examine the mouth last.

15. References

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	WHO case definition of confirmed neonatal tetanus.	An infant with **normal ability to suck and cry in the first 2 days** of life who **loses this ability between days 3 and 28** and **becomes rigid or has spasms**.
2	Organism and source in neonatal tetanus.	**Clostridium tetani** (anaerobic Gram-positive spore-former in soil/animal faeces); spores enter the **umbilical stump** via an unsterile cutting instrument or substances applied to the cord (cow dung, ash, herbs).
3	Why does maternal immunisation prevent neonatal tetanus?	Maternal **anti-tetanus IgG** (from TT/Td) **crosses the placenta** and neutralises toxin in the newborn.
4	Mechanism of action of tetanospasmin.	Retrograde axonal transport to the spinal cord/brainstem; cleaves **synaptobrevin** in inhibitory interneurons, blocking release of **GABA and glycine** → unopposed motor firing (rigidity, spasms) and autonomic instability.
5	Why does recovery from tetanus take weeks?	Toxin binding is **irreversible**; recovery needs **new nerve terminal sprouting**, which takes about **3-6 weeks**.
6	Usual age at presentation (UCG 2023).	**3-14 days** after birth, with irritability and **difficulty feeding (trismus)**, then rigidity and generalised spasms.
7	Classic signs of neonatal tetanus.	**Trismus** (cannot suck), **risus sardonicus**, clenched fists, generalised **rigidity**, **opisthotonus**, **spasms provoked by touch, noise or light**, baby **conscious** between spasms.
8	How do you distinguish tetanus spasms from seizures?	Tetanus: **conscious and alert** between spasms, spasms **provoked by stimuli**, trismus and rigidity. Seizures (sepsis, HIE, hypoglycaemia): altered consciousness, not stimulus-triggered, often subtle movements.
9	Two features of poor prognosis in tetanus.	**Incubation period below 7 days** and **period of onset (first symptom to first spasm) below 48 h**; also fever, tachycardia, frequent severe spasms, LBW, apnoea.
10	Is any lab test diagnostic of tetanus?	**No.** Diagnosis is **clinical**. CSF is normal; cultures of the stump are unreliable. Tests exclude mimics (glucose, calcium, CSF) and find complications.
11	TIG dose in neonatal tetanus (UCG 2023).	**Human tetanus immunoglobulin 500 IU IM**, split into **at least 2 sites**, **away from** the vaccine site. Give as early as possible (neutralises only unbound toxin).
12	Alternative if TIG is not available.	**Equine tetanus antitoxin** (e.g. 5,000 IU per older WHO protocols; check product) with test dose and **adrenaline ready**; IVIG (contains antitoxin) where available.
13	First-line antibiotic in neonatal tetanus (UCG 2023).	**Metronidazole 15 mg/kg IV loading over 60 min**, then **7.5 mg/kg 12-hourly** (under 4 weeks old; 8-hourly over 4 weeks). UCG lists 14 days.
14	Second-line antibiotic for neonatal tetanus (UCG).	**Benzylpenicillin 100,000 IU/kg every 12 h for 10-14 days**. Metronidazole is preferred because penicillin is a GABA antagonist.
15	Diazepam dose for tetanus spasms (UCG 2023).	**0.2 mg/kg IV** or **0.5 mg/kg rectally every 1-4 hours** as needed; watch for apnoea.

#	Question	Answer
16	Chlorpromazine dose in neonatal tetanus (UCG).	**1 mg/kg orally via NGT every 8 hours**, often alternated with diazepam.
17	Environment for a baby with tetanus.	**Quiet, dark, cool** room, minimal handling, **cluster care about 30 min after a sedative**, suction and bag-mask at the bedside.
18	Feeding plan in neonatal tetanus (UCG).	Start **IV fluids**; insert an NGT and start **expressed breast milk 24 h after admission in small frequent feeds**; mother expresses every 3 h.
19	Does recovery from tetanus give immunity?	**No.** Too little toxin to induce immunity: give a **DTP-containing vaccine** (UCG: first DPT dose at diagnosis, different site from TIG) and complete the schedule.
20	Td schedule for women of childbearing age.	**Td1** first contact; **Td2** at least **4 weeks** later; **Td3** at least **6 months** later; **Td4** and **Td5** each at least **1 year** later. Td2 at least **2 weeks before delivery**.
21	The six 'cleans' of delivery.	Clean **hands**, clean **perineum**, clean **delivery surface**, clean **blade**, clean **cord tie**, clean **cord care**.
22	Uganda cord care policy.	**Chlorhexidine 7.1% digluconate gel daily for 7 days**; keep the stump clean and dry, fold the nappy below it, **put nothing else on it** (UCG 2023).
23	Is neonatal tetanus notifiable in Uganda?	**Yes** (UCG 2023). Notify the district surveillance focal person; investigate the case (mother's Td, birth place, cord practices).
24	WHO global figures for neonatal tetanus deaths.	About **7,700 deaths in 2021**, an **84% reduction** from about 46,900 in 2000.
25	Main causes of death in neonatal tetanus.	**Apnoea/laryngospasm and respiratory failure**, **aspiration pneumonia**, secondary **sepsis**, autonomic instability, exhaustion and dehydration.
26	What is the MNT elimination target?	Fewer than **1 neonatal tetanus case per 1,000 live births in every district** per year.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. An 8-day-old baby born at home is brought because he stopped breastfeeding 2 days ago. He has a fixed grimace, clenched fists and his whole body stiffens when the nurse touches him. Between episodes his eyes are open and he follows faces. Temperature 37.6 C. What is the MOST likely diagnosis?

A Bacterial meningitis
B Neonatal tetanus
C Hypoxic-ischaemic encephalopathy
D Hypoglycaemic seizures
E Kernicterus

2. Which mechanism BEST explains the spasms of tetanus?

A Blockade of acetylcholine release at the neuromuscular junction
B Direct invasion of muscle by Clostridium tetani
C Antibody-mediated destruction of acetylcholine receptors
D Blockade of GABA and glycine release from inhibitory interneurons in the spinal cord and brainstem
E Cerebral cortical irritation causing epileptic discharges

3. A 2.6 kg, 9-day-old baby is admitted to a regional referral hospital with neonatal tetanus. He is having frequent spasms. Which set of initial orders is MOST appropriate (UCG 2023)?

A TIG 500 IU IM in two sites, first DPT dose in a different site, metronidazole 39 mg IV loading then 19.5 mg 12-hourly, diazepam 0.5 mg IV every 1-4 h as needed, quiet dark room
B Benzylpenicillin alone and discharge when spasms stop
C Phenobarbital 20 mg/kg and observe; antitoxin only if no improvement in 3 days
D TIG 500 IU IM into the same site as the DPT vaccine and ceftriaxone
E Tetanus toxoid only, because it will stimulate antibodies quickly

4. Thirty minutes after a dose of rectal diazepam, a baby with neonatal tetanus becomes floppy, stops breathing and his SpO2 falls to 78%. What is the MOST appropriate immediate action?

A Give another dose of diazepam to prevent further spasms
B Give TIG again
C Insert an NGT and feed
D Give chlorpromazine
E Open the airway, suction, and give bag-mask ventilation with oxygen until breathing resumes; then reduce the sedation dose

5. A 2.9 kg, 8-day-old baby with neonatal tetanus has received TIG and diazepam. The umbilical stump is moist and smelly. The ward stocks the drugs below. Which antibiotic order is MOST appropriate to eliminate the source of toxin, according to UCG 2023?

A Ceftriaxone 290 mg IV once daily for 5 days
B Gentamicin 14.5 mg IV once daily for 7 days
C Metronidazole 43.5 mg IV over 60 min, then 22 mg IV every 12 hours
D Benzylpenicillin 290,000 IU IV every 12 hours as first choice
E Ciprofloxacin 30 mg orally twice daily

6. A mother whose baby is admitted with neonatal tetanus received one Td dose at ANC 3 weeks before delivery. What is the MOST appropriate action for her before discharge?

A Give Td now and schedule the remaining doses to complete 5 doses
B No vaccine is needed; she has already had one dose
C Give TIG 500 IU to the mother
D Vaccinate only in her next pregnancy
E Give her antibiotics for tetanus

7. Which feature of the history BEST supports neonatal tetanus over neonatal sepsis in a 7-day-old with stiffness?

A Fever of 38 C
B Poor feeding
C Mother had prolonged rupture of membranes
D The baby sucked normally for the first 4 days and has spasms provoked by touch, with normal consciousness between them
E Jaundice to the abdomen

8. On day 2 of admission, a baby with neonatal tetanus is having fewer spasms on sedation. How should he be fed?

A Bottle-feed formula every 2 hours
B Breastfeed directly on demand
C Continue IV fluids and start small, frequent NGT feeds of expressed breast milk
D Nil by mouth for 2 weeks on IV fluids only
E Glucose water by cup

9. Which TWO features together predict the WORST prognosis in neonatal tetanus?

A Late onset (day 20) and long period of onset
B Presentation at a hospital and use of TIG
C Normal temperature and slow heart rate
D Good weight and mild trismus
E Incubation period below 7 days and period of onset below 48 hours

10. A baby is recovering from neonatal tetanus after 4 weeks in hospital. The mother asks whether he is now protected against tetanus for life. What is the BEST answer?

A Yes, natural infection gives lifelong immunity
B No; the illness does not give immunity, so he must complete his DTP-containing vaccines on schedule
C Yes, the TIG he received will protect him for life
D He only needs one booster at school age
E Immunity depends on his blood group

11. You are asked to do a lumbar puncture on a baby with suspected neonatal tetanus because a colleague is worried about meningitis. The baby is alert between frequent spasms and the fontanelle is flat. What is the MOST appropriate approach?

A Do the LP immediately without sedation
B LP is contraindicated in all newborns
C LP is usually unnecessary when the picture is classic; if meningitis cannot be excluded, sedate first and cluster the procedure, expecting normal CSF in tetanus
D Give TIG intrathecally during the LP as routine care
E Delay all treatment until CSF results are back

12. A traditional birth attendant asks what she should put on the cord after cutting it. What is the BEST advice according to Uganda's guidelines?

A Cow ghee to keep it soft
B Ash to dry it quickly
C Herbal paste to prevent bleeding
D Methylated spirit and a tight bandage
E Chlorhexidine 7.1% gel daily for 7 days, keep it clean and dry, and put nothing else on it

13. Which statement about the WHO surveillance definition of neonatal tetanus is CORRECT?

A The baby must have had normal sucking and crying in the first 2 days and lost them between days 3 and 28, with rigidity or spasms
B A positive umbilical culture for C. tetani is required
C The baby must be febrile
D Onset must be in the first 48 hours of life
E The mother must have had tetanus

14. A HC III midwife calls about a 6-day-old with trismus and spasms. The hospital is 3 hours away. TIG is not stocked. What is the MOST appropriate action?

A Keep the baby at HC III and give oral feeds
B Wait until the morning to arrange transport
C Give tetanus toxoid only and send home
D Give a sedative dose (e.g. rectal diazepam 0.5 mg/kg), a first dose of antibiotic, keep warm and quiet, and refer immediately with the mother and bag-mask on the journey
E Give a large dose of IV diazepam and send by motorcycle

15. Which is the MOST effective single public-health intervention to prevent neonatal tetanus?

A Giving antibiotics to all newborns
B Immunising women of childbearing age with tetanus-diphtheria (Td) vaccine
C Giving TIG to every newborn
D Bathing newborns immediately after birth
E Applying gentian violet to every cord

Answers & explanations

1. B · Neonatal tetanus

B is right: a home-born baby who fed normally, then stopped sucking (trismus) on day 6, with risus sardonicus, clenched fists, stimulus-provoked spasms and preserved consciousness has neonatal tetanus. A is wrong: meningitis causes reduced consciousness, fever and often a bulging fontanelle; seizures are not triggered by touch. C is wrong: HIE is abnormal from birth after a hypoxic delivery. D is wrong: hypoglycaemic seizures come with lethargy and respond to glucose. E is wrong: kernicterus occurs with deep jaundice and lethargy.

2. D · Blockade of GABA and glycine release from inhibitory interneurons in the spinal cord and brainstem

D is right: tetanospasmin travels retrogradely to the CNS and cleaves synaptobrevin in inhibitory interneurons, blocking GABA and glycine release; motor neurons fire unopposed. A is wrong: that is botulinum toxin, which causes flaccid paralysis. B is wrong: the bacteria stay at the wound; the disease is toxin-mediated. C is wrong: that is myasthenia gravis. E is wrong: the cortex is not affected, which is why the baby stays conscious.

3. A · TIG 500 IU IM in two sites, first DPT dose in a different site, metronidazole 39 mg IV loading then 19.5 mg 12-hourly, diazepam 0.5 mg IV every 1-4 h as needed, quiet dark room

A is right: UCG 2023 advises TIG 500 IU IM in at least 2 sites, a first DPT dose at a different site, metronidazole 15 mg/kg loading (39 mg) then 7.5 mg/kg 12-hourly (19.5 mg) for infants under 4 weeks, diazepam 0.2 mg/kg IV (0.52 mg) every 1-4 h, and a quiet dark room. B is wrong: penicillin is second line and antitoxin and sedation are essential. C is wrong: TIG must be given early because it only neutralises unbound toxin. D is wrong: TIG and vaccine must be given at different sites, and ceftriaxone does not replace metronidazole. E is wrong: vaccine alone takes weeks to work and does not treat the disease.

4. E · Open the airway, suction, and give bag-mask ventilation with oxygen until breathing resumes; then reduce the sedation dose

E is right: this is apnoea from over-sedation (benzodiazepine respiratory depression). Support the airway and breathing with bag-mask ventilation, then titrate sedation down. Always keep a bag and mask at the bedside. A and D are wrong: more sedatives deepen respiratory depression. B is wrong: TIG is given once and does not reverse apnoea. C is wrong: feeding a non-breathing baby risks aspiration and delays resuscitation.

5. C · Metronidazole 43.5 mg IV over 60 min, then 22 mg IV every 12 hours

C is right: UCG lists metronidazole as first line: 15 mg/kg loading over 60 min (43.5 mg), then 7.5 mg/kg (about 22 mg) every 12 hours for infants under 4 weeks; it kills C. tetani and is associated with lower mortality than penicillin. D is wrong: benzylpenicillin 100,000 IU/kg 12-hourly is the second-line option, partly because penicillin is a GABA antagonist. A is wrong: ceftriaxone is not the recommended treatment for C. tetani (though it may be used for secondary sepsis). B is wrong: gentamicin has no activity against anaerobes such as C. tetani. E is wrong: ciprofloxacin has no role here.

6. A · Give Td now and schedule the remaining doses to complete 5 doses

A is right: she had only one dose (Td1 gives no protection) and future babies are at risk; give Td2 now (at least 4 weeks after Td1) and plan Td3-Td5. B is wrong: one dose is not protective. C is wrong: TIG is passive treatment for an infected person, not prevention for her. D is wrong: delaying leaves her and future babies unprotected; Td2 should be given at least 2 weeks before a delivery. E is wrong: she does not have tetanus.

7. D · The baby sucked normally for the first 4 days and has spasms provoked by touch, with normal consciousness between them

D is right: normal sucking early, then loss of sucking with stimulus-provoked spasms and preserved consciousness is the classic tetanus picture (WHO case definition). A and B are wrong: fever and poor feeding are non-specific and common in sepsis. C is wrong: prolonged ROM is a risk factor for early-onset sepsis, not tetanus. E is wrong: jaundice is non-specific.

8. C · Continue IV fluids and start small, frequent NGT feeds of expressed breast milk

C is right: UCG advises IV fluids initially, then NGT feeds of expressed breast milk from 24 h after admission in small, frequent feeds, reducing IV fluids as feeds increase. A and B are wrong: trismus prevents sucking and oral feeding risks aspiration and spasms. D is wrong: prolonged nil by mouth causes malnutrition and infection risk. E is wrong: cup feeding a baby with trismus risks aspiration and glucose water gives no nutrition.

9. E · Incubation period below 7 days and period of onset below 48 hours

E is right: a short incubation (below 7 days) and a short period of onset (first symptom to first spasm below 48 h) indicate a large toxin load and predict severe disease and death. A is wrong: later onset and slow progression are better. B is wrong: early hospital care and TIG improve outcome. C and D are wrong: these are favourable features.

10. B · No; the illness does not give immunity, so he must complete his DTP-containing vaccines on schedule

B is right: the tiny amount of toxin that causes disease does not induce immunity, so survivors must be actively immunised (pentavalent at 6, 10, 14 weeks and boosters per UNEPI). A is wrong for the same reason. C is wrong: TIG is passive and lasts only weeks. D is wrong: he needs the full primary series. E is wrong: blood group is irrelevant.

11. C · LP is usually unnecessary when the picture is classic; if meningitis cannot be excluded, sedate first and cluster the procedure, expecting normal CSF in tetanus

C is right: tetanus is a clinical diagnosis; CSF is normal. LP is only needed if meningitis cannot be excluded, and procedures should be done after sedation and clustered, because stimulation triggers spasms and apnoea. A is wrong: an unsedated LP will provoke severe spasms. B is wrong: LP is an important test in neonates when meningitis is suspected. D is wrong: intrathecal TIG has been studied but is not routine care. E is wrong: TIG, metronidazole and sedation should not wait.

12. E · Chlorhexidine 7.1% gel daily for 7 days, keep it clean and dry, and put nothing else on it

E is right: UCG advises chlorhexidine gel daily for 7 days (if unavailable, put nothing), keep the stump clean and dry, fold the nappy below it, and do not apply anything else or bandage. A, B and C are wrong: ghee, ash and herbs can carry tetanus spores and bacteria. D is wrong: bandaging the stump is discouraged and spirit is not the national recommendation.

13. A · The baby must have had normal sucking and crying in the first 2 days and lost them between days 3 and 28, with rigidity or spasms

A is right: that is the WHO confirmed-case definition. B is wrong: diagnosis is clinical; cultures are unreliable and not required. C is wrong: fever is not part of the definition. D is wrong: onset in the first 2 days argues against tetanus (normal feeding in the first 2 days is required). E is wrong: the mother is usually unimmunised, not infected.

14. D · Give a sedative dose (e.g. rectal diazepam 0.5 mg/kg), a first dose of antibiotic, keep warm and quiet, and refer immediately with the mother and bag-mask on the journey

D is right: UCG says refer to hospital immediately; stabilise first with sedation (rectal diazepam 0.5 mg/kg works without IV access), an antibiotic dose (metronidazole or benzylpenicillin), warmth and a calm environment, and send with a health worker able to bag-ventilate. A is wrong: oral feeding risks aspiration and the baby needs hospital care. B is wrong: delay increases mortality. C is wrong: toxoid alone does not treat the disease. E is wrong: large doses of IV diazepam without monitoring cause apnoea, and an unaccompanied motorcycle trip is unsafe.

15. B · Immunising women of childbearing age with tetanus-diphtheria (Td) vaccine

B is right: maternal Td immunisation (with clean delivery and cord care) is the main reason global neonatal tetanus deaths fell by 84% between 2000 and 2021 (WHO); maternal antibodies cross the placenta. A is wrong: antibiotics do not prevent toxin disease and promote resistance. C is wrong: universal TIG is impractical and unnecessary when mothers are immunised. D is wrong: early bathing causes hypothermia. E is wrong: chlorhexidine (not gentian violet) is the recommended cord antiseptic, and cord care alone does not replace immunisation.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: A stiff baby from an island in Lake Victoria

A 10-day-old boy from Buvuma island is brought by boat to a regional referral hospital. He was born at home with a traditional birth attendant; the cord was cut with a new razor blade but dressed with cow dung 'to make it fall off quickly'. He breastfed well until day 6, then 'refused the breast'. Since yesterday his body becomes stiff and arches when touched. The mother, aged 18, attended ANC once and does not remember any injection. Weight 2.7 kg, T 37.9 C, HR 170, RR 56, SpO2 93% in air.

1. What is your diagnosis and how does it fit the WHO definition?

> **Neonatal tetanus.** He sucked and cried normally in the first 2 days, lost the ability to suck on day 6 (between days 3 and 28), and now has rigidity and spasms. Risk factors: home birth, **cow dung on the cord**, **unimmunised mother**.

2. What will you look for on examination?

> Observe before touching: **trismus**, **risus sardonicus**, clenched fists, rigidity, **opisthotonus**, spasms provoked by touch/noise/light, **consciousness between spasms**, fontanelle (normal), umbilicus (infection), hydration, chest (aspiration), vitals and SpO2.

3. What are your immediate actions in the first 30 minutes?

> Airway: suction, side position, **oxygen** to SpO2 above 90%; **bag and mask at the bedside**. Control spasms: **diazepam 0.5 mg/kg rectal = 1.35 mg** or **0.2 mg/kg IV = 0.54 mg** once IV access is obtained after sedation. **Quiet, dark room**. Glucose check. IV fluids.

4. Write the specific treatment with doses (2.7 kg, 10 days old).

> **TIG 500 IU IM**, divided between both thighs, away from the vaccine; **first pentavalent (DTP-containing) dose** in another site; **metronidazole 40.5 mg IV over 60 min**, then **20 mg IV 12-hourly** (UCG; course per local protocol, UCG lists 14 days); **diazepam 0.54 mg IV every 1-4 h** as needed alternating with **chlorpromazine 2.7 mg via NGT 8-hourly**; clean and debride the stump, chlorhexidine; antibiotics for omphalitis (cloxacillin + gentamicin) if there is spreading redness.

5. Describe the nursing and feeding plan.

> Minimal handling, **cluster care 30 min after sedation**; suction as needed; turn 2-3-hourly; monitor spasms/hour, RR, HR, SpO2, temperature, intake/output, daily weight. **IV fluids first**, then **NGT EBM from 24 h** in small, frequent feeds, increasing daily; mother expresses every 3 h; eye and mouth care; slow weaning of sedation over weeks as spasms settle.

6. On day 4 he has an apnoea with cyanosis during a spasm. How do you respond and adjust care?

> **Bag-mask ventilation with oxygen** until breathing resumes; suction; check SpO2 and glucose. Review sedation: frequent spasms with apnoea may need more regular dosing (or an infusion with monitoring), whereas apnoea after a dose suggests **over-sedation**: reduce. Consider referral for ventilation if apnoeas recur and ICU is available.

7. What public health actions are required?

> **Notify** the district surveillance focal person (notifiable disease); **case investigation** (place of birth, TBA practices, mother's Td); give the **mother Td now** and schedule Td2-Td5; community sensitisation on clean cord care and facility delivery; check Td coverage in the area.

> **Take-away:** Neonatal tetanus is diagnosed at the bedside: a baby who fed normally, then develops trismus and stimulus-provoked spasms while conscious. Treat with TIG, metronidazole, careful sedation and excellent nursing, and treat the community by vaccinating mothers.

Case 2: Spasms or seizures?

A 5-day-old girl is brought to Mengo emergency with 'fits'. She was born at Mengo by SVD after ROM for 30 hours; mother had fever in labour. She breastfed for the first 2 days, then became sleepy. Today she has rhythmic jerking of the right arm and leg lasting 2 minutes, and between episodes she is drowsy and does not open her eyes. T 38.6 C, fontanelle bulging, umbilicus clean and dry (chlorhexidine applied). Mother received 3 doses of Td.

1. Is this tetanus? Give your reasoning.

> **Unlikely.** She is **drowsy between episodes** (tetanus babies are alert), the movements are **rhythmic clonic seizures**, not stimulus-provoked spasms; there is **fever** and a **bulging fontanelle**; facility birth with clean cord care; mother **immunised (3 doses of Td)**. Most likely **early-onset sepsis with meningitis** (ROM 30 h, maternal fever).

2. What are your immediate actions?

> ABC, oxygen; **glucose** check; **phenobarbital 20 mg/kg IV** for seizures; blood culture; **ampicillin 100 mg/kg 12-hourly + gentamicin 5 mg/kg daily** (first week; meningitis doses) or ceftriaxone/cefotaxime per protocol; **LP** when stable.

3. Which features would make you reconsider tetanus?

> Home birth with **dirty cord practices**, **unimmunised mother**, **trismus**, **risus sardonicus**, **spasms provoked by touch/noise**, **normal consciousness** between spasms, normal fontanelle and CSF.

4. What is the expected CSF in each condition?

> Meningitis: **raised white cells**, high protein, low glucose, organisms on Gram stain. Tetanus: **normal CSF**.

5. What is the learning point for history-taking?

> Always ask **'Did the baby suck normally after birth, and when did it stop?'**, **'Is the baby awake between the attacks?'**, **'What was put on the cord?'** and **'How many tetanus injections did the mother receive?'**

> **Take-away:** Tetanus spasms happen in an alert baby and are provoked by stimuli; seizures with drowsiness, fever and a bulging fontanelle point to meningitis.

Case 3: Prevention at the antenatal clinic

You are with the midwife at a HC IV ANC clinic in Mubende. A 17-year-old primigravida at 30 weeks attends ANC for the first time. She has no Td card. She plans to deliver at home with her grandmother 'like everyone in our village'. Her older sister lost a baby at 9 days old with 'stiffness'.

1. What is her risk, and what probably happened to her sister's baby?

> She is **unimmunised**, planning a **home birth**: high risk of **maternal and neonatal tetanus**. Her sister's baby likely died of **neonatal tetanus** (stiffness at day 9).

2. What Td doses will you give and when?

> **Td1 today** (first contact); **Td2 at least 4 weeks later**, i.e. at about 34 weeks, at least **2 weeks before delivery**; then **Td3** 6 months after Td2, **Td4** and **Td5** a year apart.

3. What will you counsel about birth and cord care?

> Birth plan for **facility delivery** with a skilled attendant; if birth happens at home, the **six cleans** (hands, perineum, surface, blade, tie, cord care); **chlorhexidine 7.1% gel daily for 7 days**, nothing else on the cord; danger signs in the baby (stops sucking, stiffness, spasms, fever) → come immediately.

4. What else should her first ANC visit include?

> HIV, syphilis and HBsAg testing, Hb, blood group, urinalysis, BP, IPTp-SP, iron/folic acid, ITN, nutrition, deworming per guidelines, birth preparedness and complication readiness, adolescent-friendly counselling and support, family planning after birth.

5. How does this link to Uganda's MNT elimination status?

> Elimination (below 1 case per 1,000 live births in every district) must be **maintained** by high Td coverage, clean births and surveillance; every missed woman like her is a gap. Report any neonatal tetanus case for investigation.

> **Take-away:** Neonatal tetanus is prevented at ANC: give Td at first contact and the second dose at least 2 weeks before birth, plan a clean facility delivery, and teach chlorhexidine-only cord care.

References

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.